MARKETING ANALYTICS · CASE STUDY

Turning one chocolate dataset into a premium-brand strategy

An end-to-end analytics project for a premium chocolate manufacturer — segmenting consumers, mapping brand positioning, and modelling what actually drives purchase, then translating it all into a commercial strategy.

University of Leeds · MSc Marketing | Module LUBS5403M | Tool: R Studio | Individual report

4	81.2%	61.7%	9,835
Consumer segments identified	Choice-model accuracy	Variance in 2 components	Transactions analysed

BRIEF

The problem

A premium chocolate brand wanted to market more efficiently in a crowded, price-sensitive market — but didn't know *who* its most valuable customers were, *how* it was positioned against rivals, or *what* actually drove people to buy.

My task: use the company's own consumer, brand-rating and transaction data to answer those three questions and convert the findings into decisions about targeting, positioning and product.

APPROACH

Five analyses in R

I worked across three strategic themes — customer heterogeneity, customer dynamics and sustainability competitiveness — using a different technique for each question and cross-validating results so the recommendations were robust rather than reliant on a single method.

Cluster analysis (model-based · hierarchical · k-means)

PCA / factor analysis

Logistic regression

Choice-based conjoint

Market-basket / association rules

RESULT 1

Who to target — 4 segments

Three clustering methods converged on a **four-segment solution** (selected on the lowest BIC). Each segment has a distinct profile and a clear role in the strategy:

High consumption & income

Primary

The most financially attractive group. Lead with taste, quality, texture and indulgence.

Sustainability-focused

Secondary

Highest sustainability score. Win with Fairtrade, ethical sourcing and eco-friendly packaging.

Young, social-led

Tertiary

High consumption, lower spend. Reach via social, limited-edition flavours and affordable packs.

Price-sensitive

Niche

Positive on sustainability but income-constrained. Use trial packs and entry-level premium lines.

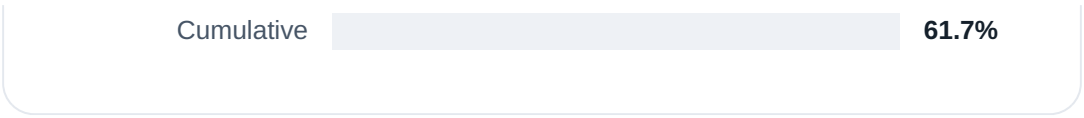
RESULT 2

How to position — the white space

PCA on brand attributes reduced the field to two interpretable dimensions capturing **61.7%** of total variation, producing a positioning map of the premium market.

Variance captured by principal components

Two components are enough to summarise how brands differ.

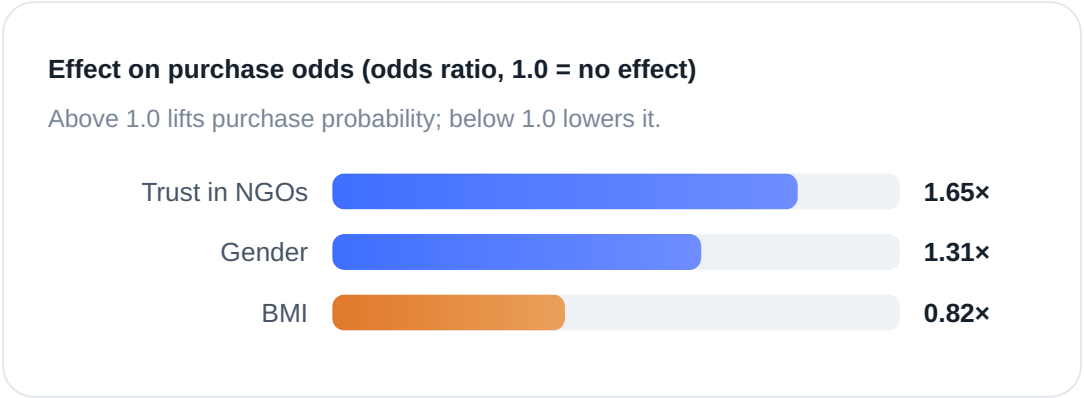


□ The map revealed an **untapped low-organic, low-sugar space**. Recommendation: avoid the crowded centre and own a differentiated **luxury-organic** position — the "Hermès of chocolate".

RESULT 3

What drives purchase

A logistic regression on a binary purchase outcome reached **81.2% accuracy** and surfaced the strongest behavioural drivers via odds ratios:

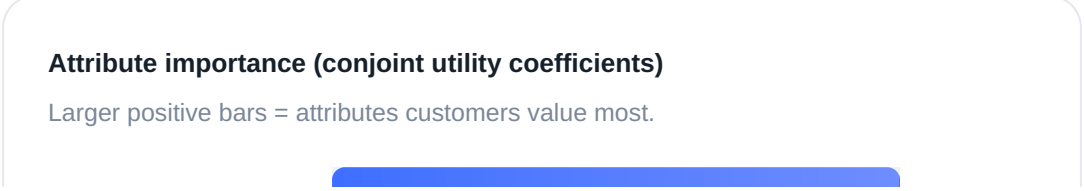


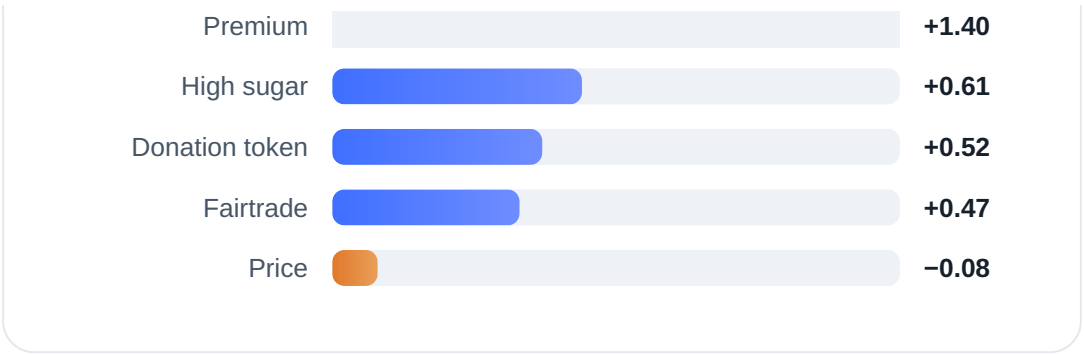
⚖ **Analytical honesty:** despite 81% headline accuracy, sensitivity was low (the class balance inflates accuracy). I flagged the model as an **early-screening tool for understanding drivers**, not a final targeting list — and recommended adding purchase-history data next.

RESULT 4

What to build — conjoint

Choice-based conjoint quantified how much each attribute moves customer choice. The full model significantly out-performed the baseline (likelihood-ratio $\chi^2 = 1000.7$, $p < 0.001$).





Willingness-to-pay analysis showed customers would pay roughly **+£4.78 for organic** and **+£5.03 for high-energy** upgrades — confirming that ethical and functional value, not country of manufacture, justify a premium price.

RESULT 5

What sells together

Market-basket analysis across **9,835 transactions** (7,004 unique baskets) used association rules to find cross-sell opportunities — chocolate clustering with snacks and dairy (crisps, nuts, yogurt, biscuits) — pointing to premium bundles and shelf-placement tests rather than discounting.

SO WHAT

The commercial recommendation

1. **Target two segments, two ways:** lead with indulgence and quality for the high-value segment; layer Fairtrade and sustainability messaging for the ethical segment.
2. **Own the luxury-organic white space** identified on the positioning map — differentiate away from the crowded centre.
3. **Launch a premium organic Fairtrade bar** with a donation element — the exact attribute mix conjoint showed drives choice — and justify the price through packaging, certification and storytelling.
4. **Test premium bundles** near snacks and gifting zones, and add purchase-history data to sharpen future targeting.

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Tools: R Studio · Excel · Full report available on request